

Review Section (Midterm)

OLS / MLE

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GOV 2003

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Today's Agenda: Review

- OLS
 - Projection Matrix
 - Annihilator Matrix
 - FWL Theorem
 - Variance of OLS
 - Asymptotics of OLS
- MLE
 - Review of Theories
 - Logistic regression
 - Bradley-Terry models

OLS and Orthogonal Projection

- Consider the linear model $\mathbf{Y} = \mathbf{X}\beta + \epsilon$. Recall that OLS estimator of β is given by

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

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OLS and Orthogonal Projection

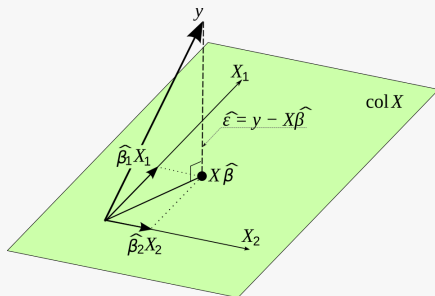
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$$MY = (I_n - X(X^\top X)^{-1}X^\top)Y = Y - X(X^\top X)^{-1}X^\top Y = Y - \hat{Y}$$

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- Annihilator matrix is a projection onto orthogonal complement of the column space of $\mathbf{X}$
- We can also show that annihilator matrix is symmetric ($\mathbf{M}^\top = \mathbf{M}$) and idempotent $\mathbf{M}^2 = \mathbf{M}$
 - That is, annihilator matrix is a projection onto orthogonal complement of the column space

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- Orthogonality 3: $\mathbf{x}^\top \hat{\boldsymbol{\epsilon}} = 0$ for any $\mathbf{x} \in S(\mathbf{X})$

- Proof: Recall the definition of column space: $\mathbf{x} = \mathbf{X} \mathbf{a}$ for some $\mathbf{a}$.

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- Zero mean: $\sum_{i=1}^n \hat{\epsilon}_i = 0$:

- As $\mathbf{X}$ includes intercept, there exists $\mathbf{a}$ such that $\mathbf{1} = \mathbf{X} \mathbf{a}$. As a result, by the previous orthogonality 3, we have zero mean.

Unbiasedness of OLS

Theorem

Unbiasedness of OLS Estimator Assume the linearity: $\mathbf{Y} = \mathbf{X}\beta + \epsilon$ with exogeneity $\mathbb{E}[\epsilon \mid \mathbf{X}] = 0$. Then,

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- **Proof**

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}] \\ &= \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \epsilon)] \\ &= \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon] \\ &= \beta + \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \epsilon] \\ &= \beta + \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}[\epsilon \mid \mathbf{X}]] = \beta\end{aligned}$$

Variance of OLS

- Now,

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon$$

and thus

$$\mathbb{V}[\hat{\beta} \mid \mathbf{X}] = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbb{V}[\epsilon \mid \mathbf{X}] \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}$$

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- Thus,

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- This is actually biased.
 - To see that, use $\text{tr}(\epsilon^\top \mathbf{M}\epsilon) = \text{tr}(\mathbf{M}\epsilon\epsilon^\top)$

$$\mathbb{E}[\hat{\sigma}^2 \mid \mathbf{X}] = \frac{1}{n} \text{tr}(\mathbb{E}[\mathbf{M}\epsilon\epsilon^\top \mid \mathbf{X}]) = \frac{1}{n} \text{tr}(\mathbf{M} \underbrace{\mathbb{E}[\epsilon\epsilon^\top \mid \mathbf{X}]}_{\sigma^2 \mathbf{I}_n}) = \frac{n-p}{n} \sigma^2$$

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- Thus, the unbiased estimator of variance is given by

$$s^2 = \frac{1}{n-p} \sum_{i=1}^n \hat{\epsilon}_i^2 = \frac{1}{n-p} \epsilon^\top \epsilon$$

Partitioned Matrix

- Sometimes, partitioned matrix can help us formulate the proof / computation
 - Let $\mathbf{A}_{11} \in \mathbb{R}^{m \times m}$, $\mathbf{A}_{12} \in \mathbb{R}^{m \times n}$, $\mathbf{A}_{21} \in \mathbb{R}^{n \times m}$, and $\mathbf{A}_{22} \in \mathbb{R}^{n \times n}$.
Then, you can write matrix as $\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}$
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Theorem (Inverse of Partitioned Matrix)

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}_{11}^{-1} + \mathbf{B}_{12} \mathbf{B}_{22}^{-1} \mathbf{B}_{21} & -\mathbf{B}_{12} \mathbf{B}_{22}^{-1} \\ -\mathbf{B}_{22}^{-1} \mathbf{B}_{21} & \mathbf{B}_{22}^{-1} \end{bmatrix}$$

where $\mathbf{B}_{12} := \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$, $\mathbf{B}_{21} := \mathbf{A}_{21} \mathbf{A}_{11}^{-1}$, and $\mathbf{B}_{22} := \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$

- We can later use this to prove partitioned regression in a different way from the last time

Partitioned Regression / FWL Theorem

- In this proof, I derive FWL theorem using partitioned matrix.

Theorem (Partitioned Regression)

Consider the partition $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ and $\beta = (\beta_1, \beta_2)$, and the regression model

$$\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \epsilon.$$

The least squared estimator $(\hat{\beta}_1, \hat{\beta}_2)$ is written as

$$\hat{\beta}_1 = (\mathbf{X}_1^\top \mathbf{M}_2 \mathbf{X}_1)^{-1} (\mathbf{X}_1^\top \mathbf{M}_2 \mathbf{Y})$$

$$\hat{\beta}_2 = (\mathbf{X}_2^\top \mathbf{M}_1 \mathbf{X}_2)^{-1} (\mathbf{X}_2^\top \mathbf{M}_1 \mathbf{Y})$$

where

$$\mathbf{M}_1 = \mathbf{I}_n - \mathbf{X}_1(\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top$$

$$\mathbf{M}_2 = \mathbf{I}_n - \mathbf{X}_2(\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top$$

Tools in Asymptotic Statistics

- **Law of Large Numbers (LLN):** If $X_1, \dots, X_n$ are i.i.d.,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mathbb{E}[X_i]$$

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- **Central Limit Theorem (CLT):** If $X_1, \dots, X_n$ are i.i.d.,

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- **Slutsky's Lemma:** If $X_n \xrightarrow{d} X$ for some random variable X and $Y_n \xrightarrow{P} c$ for some constant c ,

$$X_n Y_n \xrightarrow{d} cX$$

Asymptotics of OLS: Consistency

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- By law of large numbers,

$$\frac{1}{n} \mathbf{X}^\top \mathbf{X} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \xrightarrow{p} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]$$

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- Similarly,

$$\frac{1}{n} \mathbf{X}^\top \epsilon = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \epsilon_i \xrightarrow{p} \mathbb{E}[\mathbf{x}_i^\top \epsilon_i] = 0$$

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- Thus, by Slutsky,

$$\hat{\beta} - \beta = \left(\frac{1}{n} \mathbf{X}^\top \mathbf{X} \right)^{-1} \frac{1}{n} \mathbf{X}^\top \epsilon \xrightarrow{p} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]^{-1} \cdot 0 = 0$$

which means that $\hat{\beta}$ is consistent.

- If we use CLT for the second term, then we can derive the asymptotic normality.

MLE: Review of Basic Principle

- **Setting:** Observe data $\mathbf{Y} = (Y_1, \dots, Y_n)$ from a distribution with density (or pmf) $f_{\theta}(\mathbf{y})$, where $\theta \in \Theta$ is an unknown parameter.

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$$L(\theta) = L(\theta \mid \mathbf{y}) = f_\theta(\mathbf{y}) = \prod_{i=1}^n f_\theta(y_i) \quad (\text{under i.i.d. assumption})$$

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- **Likelihood principle:** The likelihood function contains all the information about the unknown parameter of interest.
- **MLE:** The maximum likelihood estimator is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} L(\theta)$$

Score Function

Definition (Score Function)

The score function $S_n(\mathbf{Y}, \theta)$ (or simply $S_n(\theta)$) for the data $\mathbf{Y} = (Y_1, \dots, Y_n)$ is defined as

$$S_n(\mathbf{Y}, \theta) = \frac{\partial \log f_\theta(\mathbf{Y})}{\partial \theta} = \frac{\partial \ell(\theta)}{\partial \theta}$$

where $\ell(\theta)$ is the log-likelihood function.

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- **Property:** The expected score is zero. That is, $\mathbb{E}_\theta[S_n(\mathbf{Y}, \theta)] = 0$
 - $\mathbb{E}_\theta[S_n(\mathbf{Y}, \theta)] = \int \frac{f'_\theta(\mathbf{y})}{f_\theta(\mathbf{y})} f_\theta(\mathbf{y}) d\mathbf{y} = \frac{\partial}{\partial \theta} \int f_\theta(\mathbf{y}) d\mathbf{y} = \frac{\partial}{\partial \theta} 1 = 0$

Fisher Information

Definition (Fisher Information)

The Fisher information for data $\mathbf{Y} = (Y_1, \dots, Y_n)$ is defined as

$$I(\theta) := \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f_{\theta}(\mathbf{Y}) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f_{\theta}(\mathbf{Y}) \right]$$

Fisher Information

Definition (Fisher Information)

The Fisher information for data $\mathbf{Y} = (Y_1, \dots, Y_n)$ is defined as

$$I(\theta) := \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f_{\theta}(\mathbf{Y}) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f_{\theta}(\mathbf{Y}) \right]$$

- Fisher information gives us the asymptotic variance under the correctly specified models

Theorem (Fisher's Theorem / MLE Asymptotic Normality)

Let $Y_1, \dots, Y_n \sim f_{\theta_0}$. Under the regularity conditions, the MLE estimator $\hat{\theta}_n$ achieves the asymptotic normality

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$$

where $I(\theta_0)$ is the fisher information.

Example: Logistic Regression (1)

- The likelihood function is

$$L(\beta) = \prod_{i=1}^n \pi(\mathbf{x}_i)^{Y_i} (1 - \pi(\mathbf{x}_i))^{1-Y_i}$$

and thus the logistic regression is

$$\ell(\beta) = \sum_{i=1}^n \left(Y_i \log \pi(\mathbf{x}_i) + (1 - Y_i) \log(1 - \pi(\mathbf{x}_i)) \right)$$

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- The score function is

$$\frac{\partial}{\partial \beta} \ell(\beta) = \sum_{i=1}^n \mathbf{x}_i (Y_i - \pi(\mathbf{x}_i))$$

Example: Logistic Regression (1)

- The Hessian is

$$\frac{\partial^2}{\partial \beta^2} \ell(\beta) = - \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \pi(\mathbf{x}_i) (1 - \pi(\mathbf{x}_i))$$

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- By information equality,

$$I_n(\theta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \beta^2} \ell(\beta) \right] = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \pi(\mathbf{x}_i) (1 - \pi(\mathbf{x}_i))$$

- Note that this assumes fixed design (i.e., $\mathbf{x}_i$ is non-random.)

Example 2: Bradley Terry Model (1)

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$$L(\beta) = \prod_{1 \leq j < j' \leq J} \left(\frac{e^{\beta_j}}{e^{\beta_j} + e^{\beta_{j'}}} \right)^{w_{jj'}} \left(\frac{e^{\beta_{j'}}}{e^{\beta_j} + e^{\beta_{j'}}} \right)^{w_{j'j}}$$

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- And thus the log-likelihood is

$$\ell(\beta) = \sum_{1 \leq j < j' \leq J} \left[W_{jj'} \beta_j + W_{j'j} \beta_{j'} - (W_{jj'} + W_{j'j}) \log(\theta_j + \theta_{j'}) \right].$$

where $\theta_j := e^{\beta_j}$

Example 2: Bradley-Terry Model (2)

- The score is

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{j' \neq j} \left[w_{jj'} - (w_{jj'} + w_{j'j}) \frac{\theta_j}{\theta_j + \theta_{j'}} \right].$$

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- The Hessian is, for $j \neq k$,

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = (W_{jk} + W_{kj}) \frac{\theta_j \theta_k}{(\theta_j + \theta_k)^2},$$

and for $j = k$,

$$\frac{\partial^2 \ell}{\partial \beta_j^2} = - \sum_{j' \neq j} (W_{jj'} + W_{j'j}) \frac{\theta_j \theta_{j'}}{(\theta_j + \theta_{j'})^2}.$$

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- Thus, for $j \neq k$,

$$I_n(\beta)_{jk} = -(W_{jk} + W_{kj})\frac{\theta_j\theta_k}{(\theta_j + \theta_k)^2},$$

and

$$I_n(\beta)_{jj} = \sum_{j' \neq j} (W_{jj'} + W_{j'j})\frac{\theta_j\theta_{j'}}{(\theta_j + \theta_{j'})^2}.$$